

BACKGROUND

- ❑ Extreme heat weather events linked to rising global temperatures have increased in frequency and intensity.
- ❑ New York City is known for its densely populated urban environment and continues to face worsening heat conditions.
- ❑ As a result, heat-related weather events are emerging as a significant public health concern, placing increasing strain on existing public health systems.

QUESTION

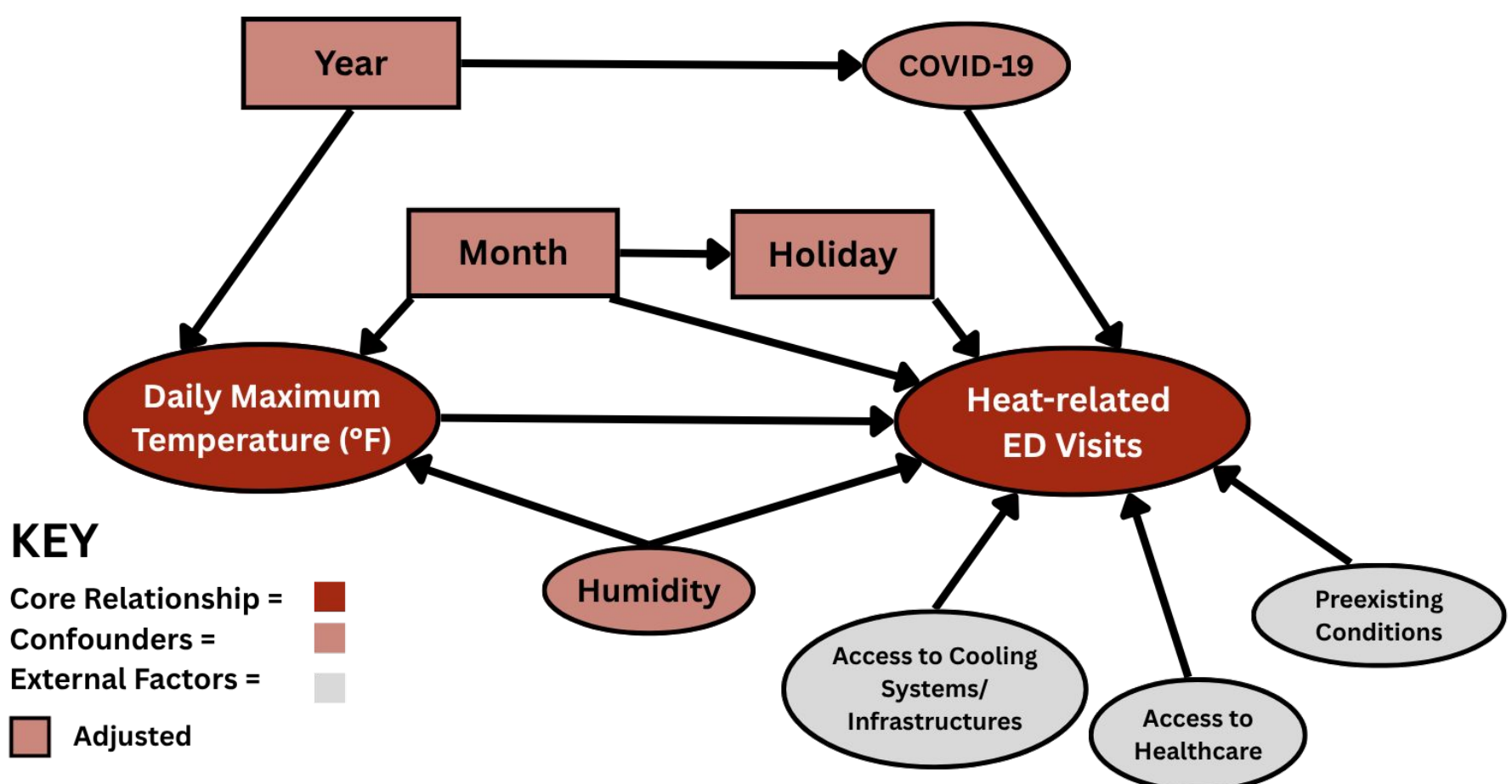
What is the association between heat-related emergency department (ED) visits, in New York City, and daily maximum temperature at lags of 0–6 days, and how can this relationship be quantified to predict future heat-related ED visits?

METHODS

Dataset

Syndromic surveillance data (2017-2026) of heat-related ED visits and maximum daily temperature measured at LaGuardia Airport and reported to the New York City Department of Health and Mental Hygiene (NYC DOHMH).

Figure 1. Direct Acyclic Graph



Estimation
Model

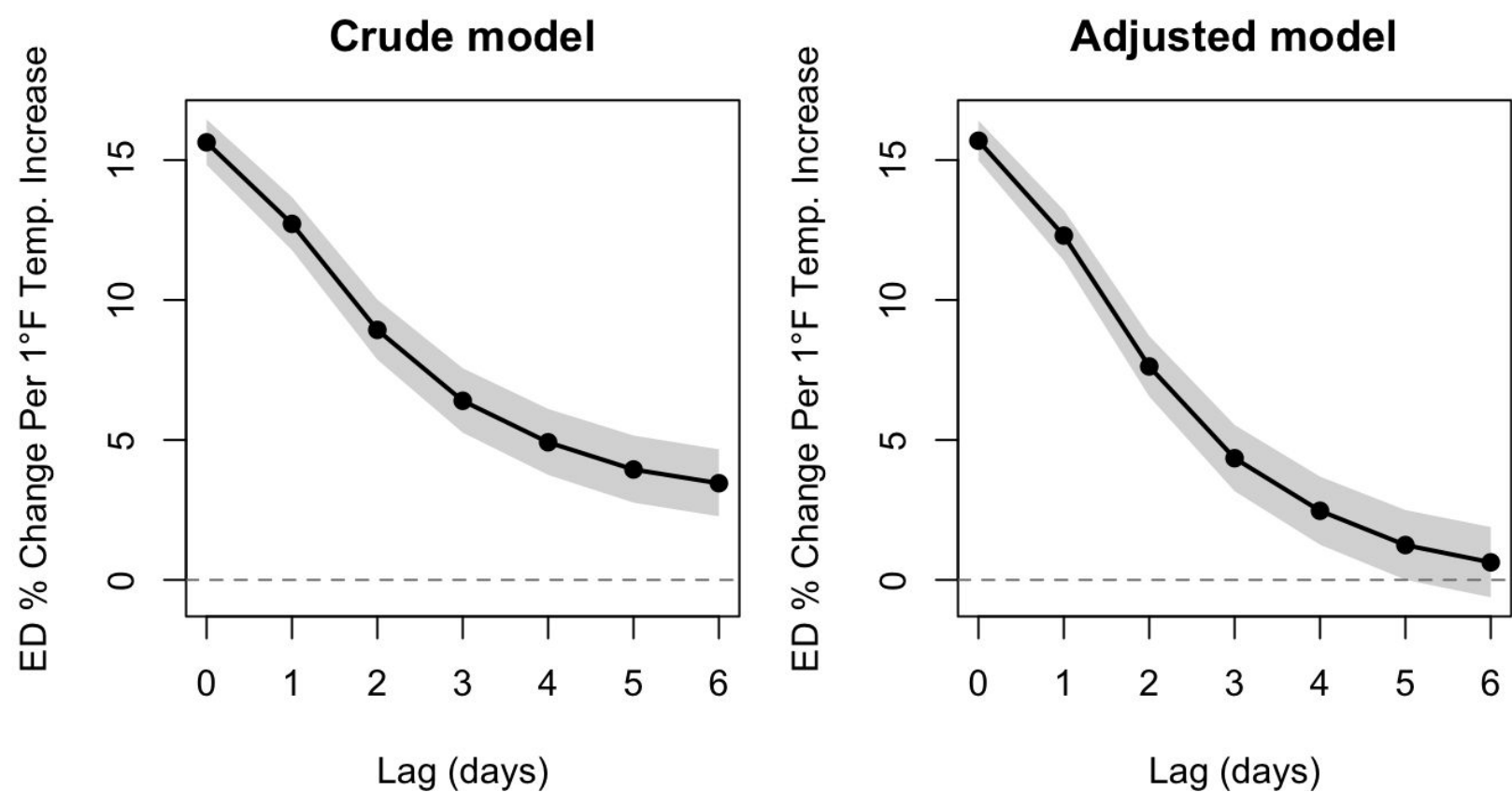
- ❑ Used a directed acyclic graph (DAG) to help identify time-varying confounders.
- ❑ Tested various approaches (Linear, Poisson, Quasi-Poisson, and Negative Binomial regression) to select final regression model.
- ❑ Created separate Quasi-Poisson regression models with all confounders fit for lag days 0-6 to evaluate the impact of each lag day on heat-related ED visits.

Prediction
Model

Quasi-Poisson regression model from 2017-2025 was utilized to predict 2026 heat-related ED visits.

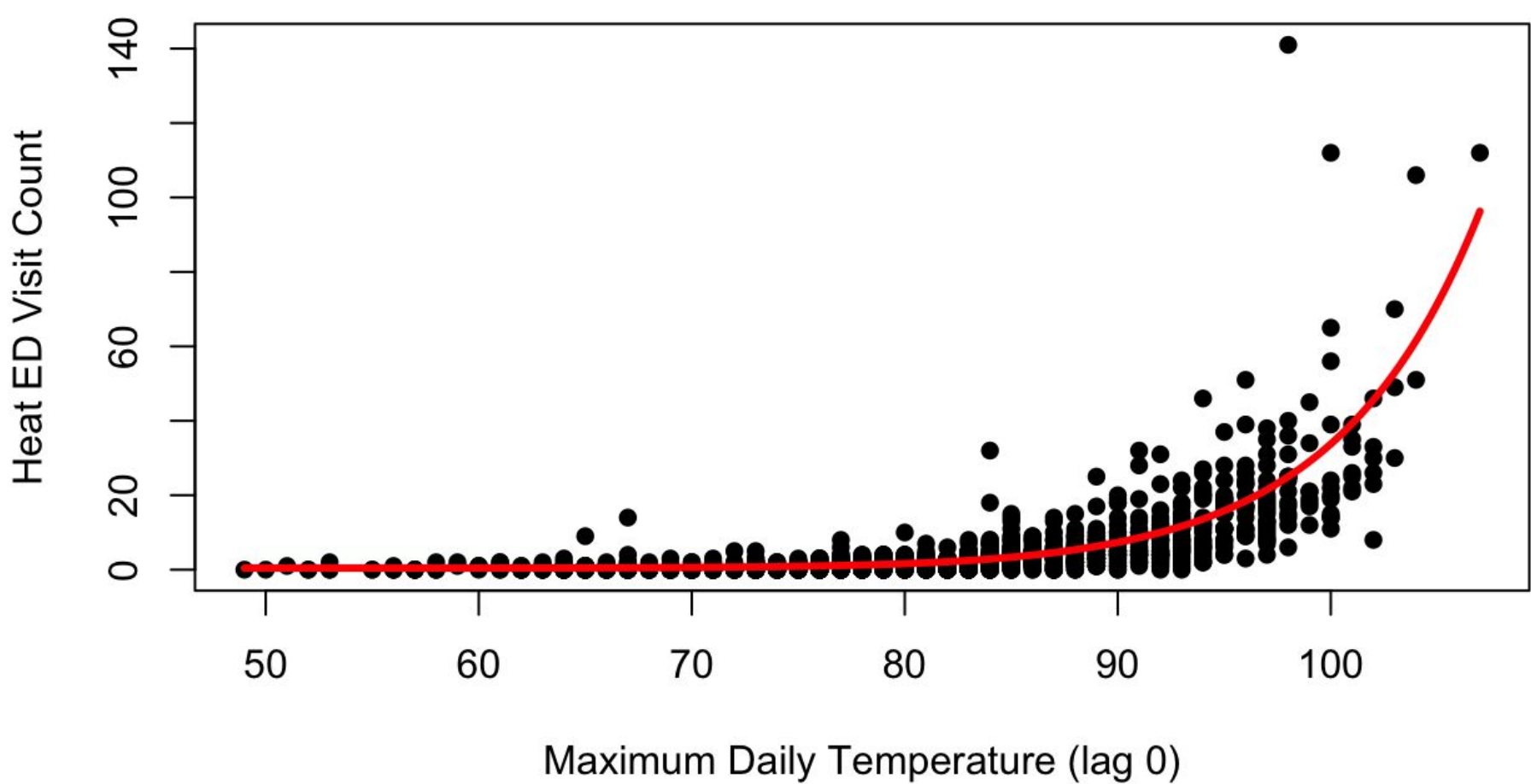
RESULTS

Figure 2. Estimation Model



Heat-related ED visits had the largest magnitude association with same-day (lag 0) daily maximum temperature, though associations with temperature on prior days remained positive and statistically significant through lag 4. Separate models were used for each lag of exposure.

Figure 3. Quasi-Poisson Regression with Natural Spline

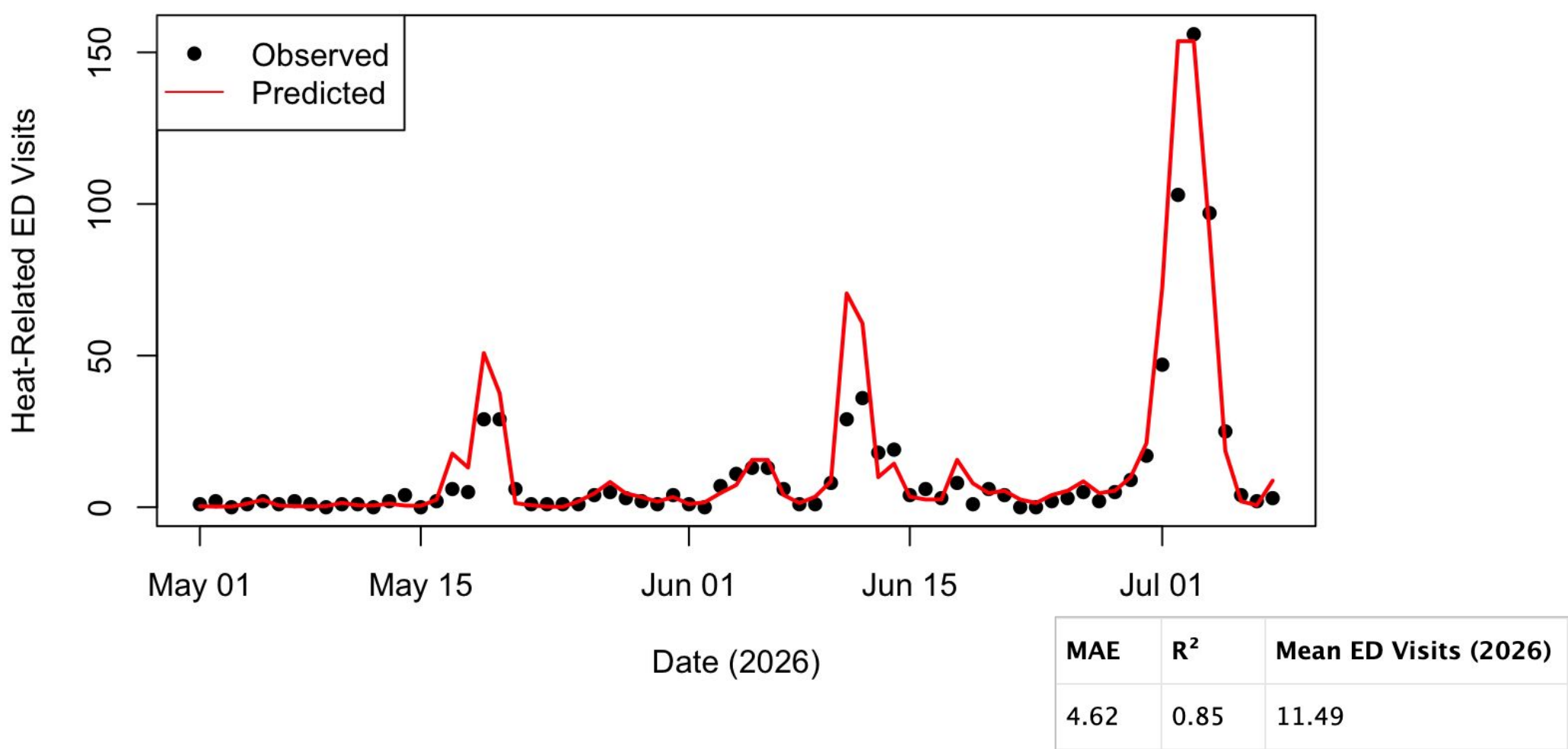


Coefficient	Estimate	Pr(> t)	95% CI
MAX_DAILY_TEMP (lag 0)	15.7%	<.001	(15.0%, 16.4%)

After adjusting for year, month, and summer holidays, we found that a 1°F increase in same-day maximum daily temperature was associated with a 15.7% increase in the expected rate of heat-related ED visits.

RESULTS CONT.

Figure 4. Observed vs. Predicted Heat-Related ED Visits for 2026



The final Quasi-Poisson regression model predicted heat-related ED visit counts for the 2026 dataset using daily maximum temperature. The model explained 85% of the variability ($R^2 = 0.85$) and had an MAE of approximately 5 ED visits.

DISCUSSION

Findings

- ❑ Statistically significant association between max daily temperature and heat-related ED visits, even after adjusting for summer holidays, month, and year.
- ❑ Same day exposure (lag 0) had the strongest association with heat-related ED visits.

Limitations

- ❑ Additional potential confounding factors, such as daily humidity and external factors including access to cooling centers and healthcare services were not accounted for in the study.
- ❑ Analysis was limited to data collected only during the months of May-September.

FUTURE DIRECTIONS

- ❑ Compute MAE as a function of temperature to look at prediction ability particularly for extreme heat days.
- ❑ Predicting ED visits can support public health preparedness through proactive interventions before extreme heat events.
- ❑ Findings can inform heat safety messaging, cooling center implementation, and healthcare system planning for increased ED demand.

